Axles and alignment

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use measurements and observations to improve straight rolling.

SUCCESS CRITERIA - I can...

1. Check axle parallelism and wheel clearance.
2. Record a repeatable alignment test.
3. Use evidence to choose one adjustment.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A 50-turn coil has flux per turn change by 0.007 Wb in 0.5 s. Find average induced-emf magnitude.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: clearance = available gap - part width.
Use positive clearance and check free rotation. Alignment and wheel rubbing are separate checks; change one feature at a time.

Worked reasoning: Parallel axles can still cause drag if wheels rub the chassis. Alignment and clearance are separate checks; a straight top view does not verify free rotation.

Your turn: Draw a top view and a side detail marking an axle direction and a wheel gap.

Marzano check - circle: ① ② ③

03 Represent and interpret

On your top-view sketch, mark both axle lines. Add a check for parallelism and show clearance between each wheel and the chassis.



Top view: the large rectangle is the chassis; small rectangles are wheels; the two crossbars are axles.

Marzano check - circle: ① ② ③

04 Solve and justify

A supplied straight-line run has forward displacement 2.0 m and lateral displacement .12 m. Find the direction angle using $\tan^{-1}(.12/2.0)$, and the displacement magnitude.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

A car veers right once. Does that prove the rear axle is crooked? Name another possible cause and a discriminating check.

Marzano check - circle: 1 2 3

06 Choose a method

A 4 Ω resistor carries 6 A. Find power using both IV and I^2R, showing the implied voltage.

Marzano check - circle: 1 2 3

07 Exit check

Compare two trial sets by mean offset and spread. Explain why one excellent run is weaker evidence than repeated improvement.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Repeated 3 m runs finish 0.20, 0.24, 0.22 m left of the line. Find mean deviation and range; design a controlled adjustment and a fair comparison after it.

Marzano check - circle: 1 2 3